

RAJ S

Nagarajan Shunmugam

AI Engineer · CRISPR & NGS Researcher · Co-Founder & CTO · DVM Candidate

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PROFILE

Nine years of applied AI across six countries: autonomous trucks at Scania, medical imaging in Berlin, veterinary medicine in California, and CRISPR machine learning in Wrocław. Three-time startup co-founder. Currently a CRISPR & NGS data researcher at the Hirsfeld Institute (Polish Academy of Sciences), CTO of MilkAI, and a veterinary medicine student — building the bridge between AI and animal health.

EXPERIENCE

- CRISPR & NGS Data Researcher** — Hirsfeld Institute of Immunology & Experimental Therapy, PAN
Wrocław, Poland
Nov 2025 — present
ML pipelines for CRISPR off-target prediction on genomic, NGS and multi-omics data (k-mer, scalar, one-hot encodings; CNN and Transformer models). Multi-agent AI systems that design and validate wet-lab experiment protocols.
- Co-Founder & CTO** — MilkAI LLC
California, USA
Jul 2025 — present
Product studio shipping production-ready web, mobile and AI-first apps — design-to-deploy with CI/CD, dozens of launches.
- Artificial Intelligence Engineer** — Veterinary Information Network
California, USA
May 2023 — Jul 2025
AI engineering for the world's largest online veterinary community.
- Co-Founder** — Zaide.ai
Remote
Jan 2023 — Aug 2024
- Mentor & Alpha Tester** — Coursera
Orange County, CA
Apr 2017 — Jun 2023
Six years mentoring the Neural Networks & Deep Learning course; alpha-tested unreleased AI courses; mentored 3D-printing applications.
- Assistant Lecturer** — Coventry University (Wrocław campus)
Wrocław, Poland
Aug 2022 — Mar 2023
Taught AI, IoT and software engineering.
- Senior AI Instructor** — Techeta Technologies
India
Sep 2021 — Sep 2022
Trained technical and corporate teams in applied AI.
- Co-Founder & CTO** — Neurons4vet.ai
Toronto, Canada
Jan 2020 — Dec 2021
First venture at the intersection of AI and veterinary medicine.
- Cohort Member** — Entrepreneur First
Berlin, Germany
Mar — Aug 2021
Europe's leading deep-tech talent investor (300+ startups, \$2.7B+ portfolio).
- Thesis Student** — Scania Group
Stockholm, Sweden
Jan 2020 — Mar 2021
Truck visual perception: classification, detection, semantic segmentation, depth estimation, reinforcement learning.
- Machine Learning Engineer** — ai4medicine
Berlin, Germany
May — Oct 2020
Computer-vision AI for stroke and bleeding detection.
- Senior Design Engineer** — KTH Formula Student
Aug 2019 — Jan 2020

Stockholm, Sweden

Autonomous race car perception; LiDAR point clouds; NVIDIA Jetson deployment.

Project Assistant — Indian Institute of Technology, Delhi

New Delhi, India

Driverless car: PID tuning, GPS+IMU+LiDAR+radar sensor fusion.

Sep 2017 — Feb 2018

Research Scholar — Indian Institute of Technology, Madras

Chennai, India

Embedded programming, biomedical research lab (bio-medical kiosks, SpO2 sensors).

Mar — Aug 2017

EDUCATION

Wrocław University of Environmental and Life Sciences (UPWR) — Veterinary

2021 — 2027

Medicine — Veterinary Surgeon (DVM), in progress

Technische Universität Berlin — M.Sc. Autonomous Systems — Robotics & AI (full scholarship)

2018 — 2021

KTH Royal Institute of Technology — M.Sc. Autonomous Systems — Robotics & AI (EIT Digital double degree)

2018 — 2021

Aalto University — EIT Digital Summer School — Disrupting Retail

2019

Anna University — B.E. Electrical & Electronics Engineering (partial scholarship)

2013 — 2017

SKILLS

AI/ML: CNNs & Transformers, deep reinforcement learning, imitation learning, multi-agent systems, data analysis ·

Genomics: CRISPR guide design, NGS & multi-omics, k-mer/one-hot encodings · **Autonomy:** detection & segmentation, depth estimation, LiDAR, sensor fusion, NVIDIA Jetson · **Engineering:** technical leadership, IoT & embedded (Arduino), 3D printing, CI/CD

PUBLICATION & HONORS

Publication: Deep Learning for Hardware-Constrained Cars — Imitation Learning & Deep Reinforcement Learning

Honors: Full scholarship (master's) · Partial scholarship (bachelor's) · **Languages:** English, Tamil (native)